

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.50)

Metric

$$ds^2 = -dt \otimes dt + t dr \otimes dr + V(t, z, r, \phi)^2 dz \otimes dz + r^2 t d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, z, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\sqrt{\frac{1}{2}} \sqrt{t} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\sqrt{\frac{1}{2}} \sqrt{t} \right) dr$$

$$m = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(i \sqrt{\frac{1}{2}} r \sqrt{t} \right) d\phi$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(-i \sqrt{\frac{1}{2}} r \sqrt{t} \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2}}{2\sqrt{t}} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\sqrt{2}}{2\sqrt{t}} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(\frac{i\sqrt{2}}{2r\sqrt{t}} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(-\frac{i\sqrt{2}}{2r\sqrt{t}} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2}}{2\sqrt{t}} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2}}{2\sqrt{t}} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z + \frac{i\sqrt{2}}{2r\sqrt{t}} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z - \frac{i\sqrt{2}}{2r\sqrt{t}} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{t}V(t, z, r, \phi)} + \frac{\sqrt{2}}{8t} - \frac{\sqrt{2}}{4r\sqrt{t}}$$

$$\sigma = \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{t}V(t, z, r, \phi)} - \frac{\sqrt{2}}{8t} + \frac{\sqrt{2}}{4r\sqrt{t}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial t}V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\sqrt{2}\frac{\partial}{\partial r}V(t, z, r, \phi)}{4\sqrt{t}V(t, z, r, \phi)} - \frac{\sqrt{2}}{8t} - \frac{\sqrt{2}}{4r\sqrt{t}}$$

$$\lambda = -\frac{\sqrt{2}\frac{\partial}{\partial t}V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\sqrt{2}\frac{\partial}{\partial r}V(t, z, r, \phi)}{4\sqrt{t}V(t, z, r, \phi)} + \frac{\sqrt{2}}{8t} + \frac{\sqrt{2}}{4r\sqrt{t}}$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{2}\frac{\partial}{\partial \phi}V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)}$$

$$\beta = \frac{i\sqrt{2}\frac{\partial}{\partial \phi}V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}}{8t}$$

$$\gamma = \frac{\sqrt{2}}{8t}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\frac{\partial^2}{(\partial t)^2}V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\frac{\partial^2}{\partial t \partial r}V(t, z, r, \phi)}{2\sqrt{t}V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t}V(t, z, r, \phi)}{8tV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2}V(t, z, r, \phi)}{4tV(t, z, r, \phi)} - \frac{\frac{\partial}{\partial r}V(t, z, r, \phi)}{4t^{\frac{3}{2}}V(t, z, r, \phi)} + \frac{1}{8t^2}$$

$$\Phi_{01} = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)} - \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rtV(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8rt^{\frac{3}{2}}V(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2tV(t, z, r, \phi)}$$

$$\Phi_{02} = \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8tV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4tV(t, z, r, \phi)}$$

$$\Phi_{10} = -\frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rtV(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8rt^{\frac{3}{2}}V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2tV(t, z, r, \phi)}$$

$$\Phi_{11} = \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8tV(t, z, r, \phi)} + \frac{1}{16t^2} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{4rtV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{4r^2tV(t, z, r, \phi)}$$

$$\Phi_{12} = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rtV(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8rt^{\frac{3}{2}}V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2tV(t, z, r, \phi)}$$

$$\Phi_{20} = \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8tV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4tV(t, z, r, \phi)}$$

$$\Phi_{21} = -\frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4r\sqrt{t}V(t, z, r, \phi)} - \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rtV(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8rt^{\frac{3}{2}}V(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2tV(t, z, r, \phi)}$$

$$\Phi_{22} = -\frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2\sqrt{t}V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8tV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4tV(t, z, r, \phi)} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{4t^{\frac{3}{2}}V(t, z, r, \phi)} + \frac{1}{8t^2}$$

$$\Lambda = \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{12tV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12tV(t, z, r, \phi)} - \frac{1}{48t^2} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{12rtV(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{12r^2tV(t, z, r, \phi)}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} + \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 \sqrt{t} V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8 t V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 t V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{4 t^{\frac{3}{2}} V(t, z, r, \phi)} - \frac{1}{8 t^2}$$

$$\Psi_1 = -\frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r \sqrt{t} V(t, z, r, \phi)} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r t V(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r t^{\frac{3}{2}} V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 t V(t, z, r, \phi)}$$

$$\Psi_2 = \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{24 t V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 t V(t, z, r, \phi)} + \frac{1}{24 t^2} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{6 r t V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{6 r^2 t V(t, z, r, \phi)}$$

$$\Psi_3 = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r \sqrt{t} V(t, z, r, \phi)} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r t V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r t^{\frac{3}{2}} V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 t V(t, z, r, \phi)}$$

$$\Psi_4 = -\frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 \sqrt{t} V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi)}{8 t V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 t V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{4 t^{\frac{3}{2}} V(t, z, r, \phi)} - \frac{1}{8 t^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"